Pre-analysis plan

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1. Variables

I will use the following variables in my analysis:

- **sumkids**: the number of children that the respondent has
- **kidsex01-kidsex19**: a set of variables indicating the sex at birth of each child that the respondent has.
- **plj0625**: a variable that records agreement with the sentence "At home, the husband should have the final say" (with higher values meaning more agreement)
- **plh0004**: variable measuring political orientation on the left-right scale (with higher values meaning more right-wing)
- **plh0303**: variable that records to what extent, according to the respondent, in the family men are involved with housework (with higher values meaning less involved)
- **gebjahr**: year of birth of the respondent
- **syear**: year in which the survey was taken
- **corigin**: country of origin of the respondent
- **sex**: gender of the respondent (1 for males, 2 for females)
- **pid**: participant ID identifying each respondent.

The variables kidsex01-kidsex19 will be used to create a binary variable equal to 1 if the respondent has at least one daughter and 0 otherwise.

All variables will first be cleaned as needed. Specifically, for some of these variables, non responses are coded as negative numbers in the SOEP raw files (with different negative numbers signalling different reasons for non responses). Whenever this is the case and the negative numbers would contaminate the analysis, they will be recoded as missing values.

For ease of interpretation, I will recode the sex variable into a "male" variable equal to 1 for males and 0 for females.

The variables plj0625, plh0004 and plh0303 will be standardized to have a mean of zero and a standard deviation of one. Moreover, the resulting standardized variable will be multiplied by -1, so that higher values indicate respectively more support for gender equality; more left-wing political orientation; more male participation in housework in the family.

2. Inclusion/exclusion criteria

I will only include in the sample individuals with at least one child (ie, sumkids>0). This is because I think that people without children are not a good comparison group for people with children: there are substantial pre-existing differences that drive the decision to have children, and at the same time having children affects people in a variety of ways, independent of the gender of the children. However, conditional on having a child, their sex at birth is randomly determined. Therefore, individuals whose children are all boys are a good comparison group for people with at least a daughter (after controlling for the number of children, which I will do by including a full set of "number of children" fixed effects).

I will include all available survey years.

3. Exact analysis

For testing H1-H3, I will estimate the following OLS regression:

$$y_{it} = \alpha + \beta D_{it} + \delta X_{it} + \varepsilon_{it}$$

Where y is the outcome of interest; *i* indexes individuals; *t* indexes the survey year; and X is a vector of control variables that includes a full set of fixed effects for the discrete variables sumkids, gebjahr, syear, corigin, and male. To be clear, this means that, for each of these control variables, I include an indicator for each possible category.

The outcome variable *y* is a standardized and sign-reversed version of: plj0625 for H1; plh0004 for H2; plh0303 for H3.

For testing H4, I will estimate the following OLS regression:

$$y_{it} = \alpha_0 + \alpha_1 D_{it} + \alpha_2 male_i + \beta D_{it} * male_i + \delta X_{it} + \varepsilon_{it}$$

Where *y* is the standardized and sign-inverted version of plj0625, and X is defined as above, except that now it does not include *male*.

In all specifications, I will use robust standard errors clustered by individual (as identified by the pid variable) and survey year (as identified by the syear variable), using two-way clustering.

4. Interpretation of results

For all four hypotheses, an estimated coefficient $\hat{\beta}$ which is positive and non-negligible in magnitude will be considered "in line with the hypothesis", at least in terms of direction of the effect. Although any threshold is necessarily arbitrary, my operational definition for "non-negligible in magnitude" will be higher than 0.025 standard deviations of the outcome. Therefore, I will consider the result "in line with the hypothesis" if $\beta > 0.025$.

Of course, the confidence we should have in the result will depend on the precision of the estimate, as measured by its estimated standard error.